

Weather & Climate

Will it rain today? Why is the coast cooler than inland? What turns a quiet sky into a thunderstorm or a hurricane? It all comes down to the Sun heating our planet unevenly — and the air and water moving that heat around.

 Georgia Standard S6E4

WHAT YOU'LL LEARN

- | | |
|---|------------------------------------|
| 01 Weather vs. Climate | 02 Layers of the Atmosphere |
| 03 Uneven Heating | 04 How Wind Forms |
| 05 Air Pressure, Fronts & Storms | 06 Hurricanes |



4.1 Weather vs. Climate

People mix up these two words all the time, but they mean different things. **Weather** is what the air is doing *right now* in one place — sunny, rainy, hot, windy. **Climate** is the *average* weather of a place over many years.

★ Easy Way to Remember

Weather is your mood; climate is your personality. Your mood can change in an afternoon (weather), but your personality is your usual pattern over a long time (climate). A single rainy day doesn't change Georgia's warm, humid climate.

	Weather	Climate
Time frame	Right now, today, this week	Averaged over 30+ years
Changes	Quickly, hour to hour	Slowly, over a long time
Example	"It's storming this afternoon."	"Summers here are hot and humid."

? Quick Check 4.1

1. Is "it snowed today" describing weather or climate?
2. In your own words, what is the difference between weather and climate?

4.2 Layers of the Atmosphere

The **atmosphere** is the blanket of gases surrounding Earth. It's mostly nitrogen (78%) and oxygen (21%), with small amounts of other gases. The atmosphere is divided into layers, each with different features.

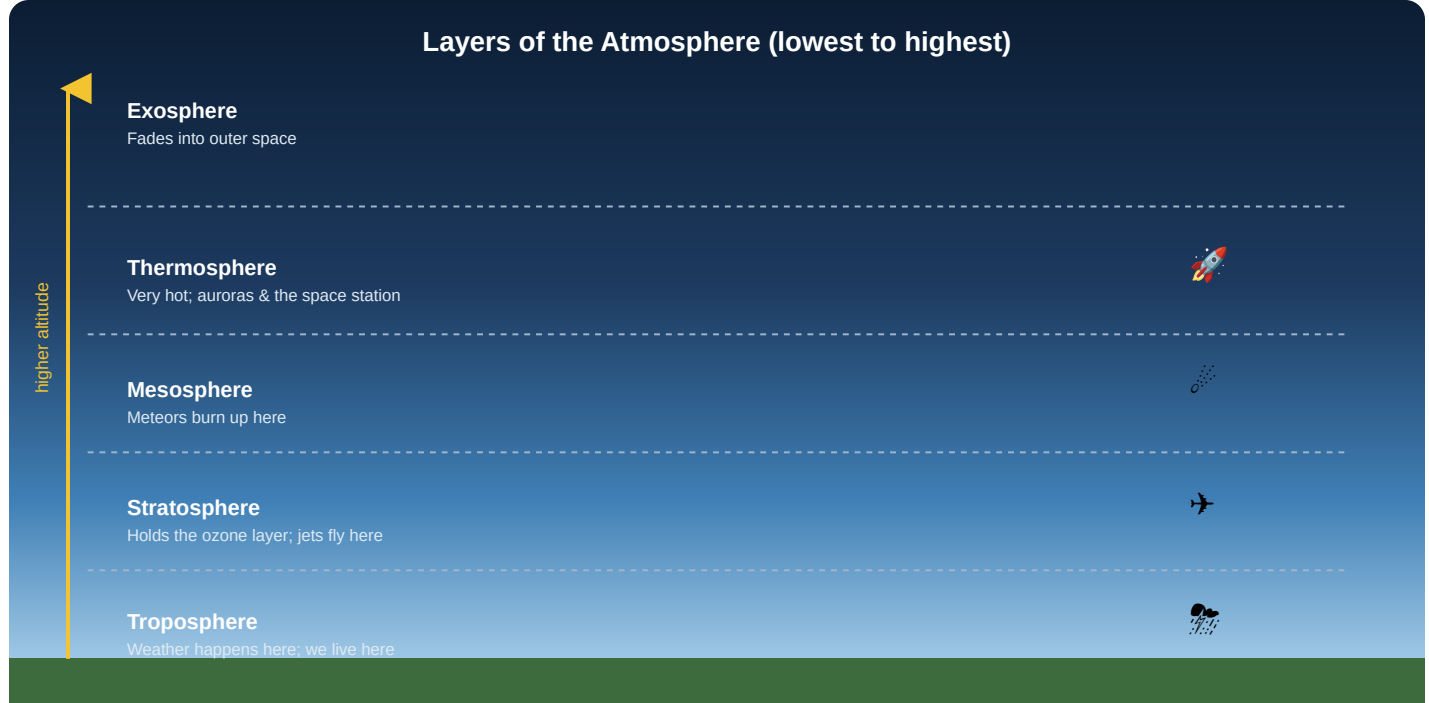


Figure 4.1 — The atmosphere's layers, from the ground up. We live in the troposphere, where all weather happens.

Layer	What's There
Troposphere	The lowest layer; where we live and where all weather happens
Stratosphere	Contains the ozone layer; jets fly here for a smoother ride
Mesosphere	Coldest layer; most meteors burn up here
Thermosphere	Very hot; auroras (northern lights) and the space station are here
Exosphere	The outermost layer; slowly fades into outer space

The Ozone Layer & Greenhouse Gases

Two parts of the atmosphere matter a lot for life. The **ozone layer** (in the stratosphere) absorbs most of the Sun's harmful ultraviolet (UV) rays, like sunscreen for the whole planet. **Greenhouse gases** (such as carbon dioxide, methane, and water vapor) trap heat near Earth's surface, keeping the planet warm enough for life.

★ The Greenhouse Effect

Greenhouse gases work like a blanket: they let sunlight in, then trap some of the heat that tries to escape back to space. Without any greenhouse gases, Earth would be a frozen ball. The concern today is that **extra** greenhouse gases are trapping **too much** heat, slowly warming the planet.

? Quick Check 4.2

1. Which layer of the atmosphere do we live in, and what happens there?
2. What does the ozone layer protect us from?
3. What do greenhouse gases do to heat?

4.3 Uneven Heating

Almost everything about weather starts with one fact: the Sun does **not** heat the Earth evenly. Different surfaces soak up and release the Sun's energy at different speeds.

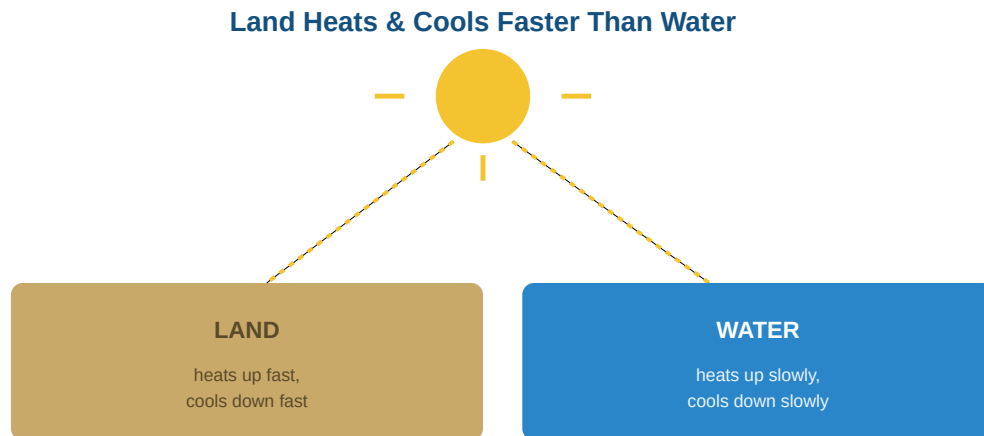


Figure 4.2 — Given the same sunshine, land heats up and cools down faster than water. Water changes temperature slowly.

This is why a sandy beach can burn your feet at noon while the ocean stays cool — and why that same sand feels cold at night while the water still feels warm. Land changes temperature quickly; water changes slowly. The Sun also heats the equator more directly than the poles, which is why it's hot near the equator and cold near the poles.

★ Why This Matters

Uneven heating creates areas of warm air and cool air sitting next to each other. As you'll see next, that temperature difference is exactly what makes the wind blow.

? Quick Check 4.3

1. Which heats up faster in the Sun: land or water?
2. Why is it hotter at the equator than at the poles?

4.4 How Wind Forms

Wind is simply moving air. But what makes air move? The answer connects straight back to uneven heating. When air gets warm, it becomes lighter and **rises**. Cooler, heavier air then flows in to take its place — and that moving air *is* the wind.

Why Wind Blows: Warm Air Rises, Cool Air Moves In

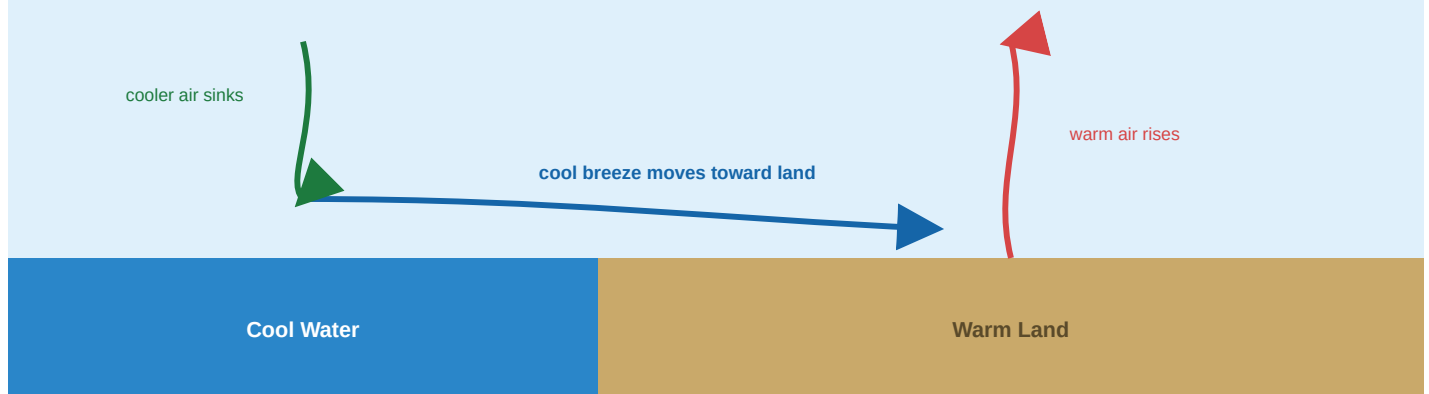


Figure 4.3 — During the day, land heats faster than water. Warm air rises over the land, and a cool breeze flows in from the water to replace it.

Local Winds and Global Winds

The same idea works at every size. A **sea breeze** at the beach is a small, local wind. On a global scale, the Sun heats the equator strongly, that warm air rises, and air from cooler regions flows toward the equator. Because Earth is also **rotating**, these global winds get bent into the steady patterns sailors have used for centuries.

Air pressure: the weight of the air pressing down. Warm rising air creates **low pressure**; cool sinking air creates **high pressure**. Wind always flows from high pressure toward low pressure.

? Quick Check 4.4

1. What happens to air when it gets warm?
2. Does wind flow from high pressure to low, or low to high?
3. What causes a sea breeze during the day?

4.5 Air Pressure, Fronts & Storms

Large bodies of air with similar temperature and moisture are called **air masses**. Some are warm, some cold; some wet, some dry. Where two different air masses meet, the boundary is called a **front** — and fronts are where most of our exciting weather happens.

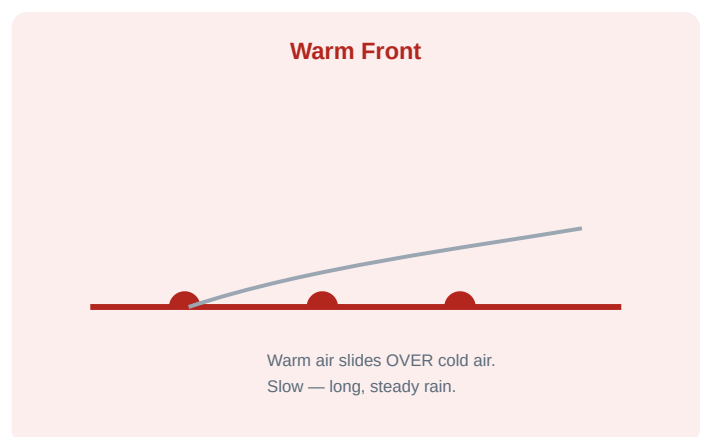
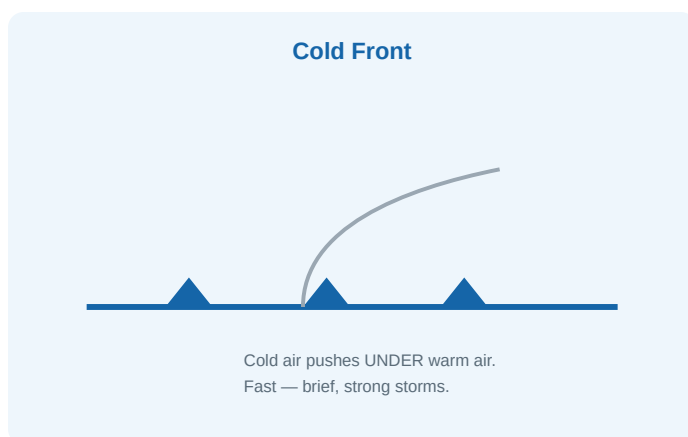


Figure 4.4 — A cold front (left): cold air shoves under warm air, often causing quick, strong storms. A warm front (right): warm air slides over cold air, bringing slow, steady rain.

Front	What Happens	Typical Weather
Cold Front	Cold air pushes under and lifts warm air quickly	Sudden, strong storms; then cooler, clearer air
Warm Front	Warm air slides slowly up and over cold air	Long periods of light, steady rain; then warmer air

Thunderstorms and Tornadoes

When warm, moist air rises very fast — often along a cold front — it can build tall storm clouds and produce a **thunderstorm** with heavy rain, lightning, and thunder. If the rising air also starts to spin, a thunderstorm can sometimes produce a **tornado**: a violently rotating column of air that reaches the ground. Georgia sees both, especially in spring.

! Pressure and Weather

Low pressure usually means stormy weather; high pressure usually means clear weather. That's because low pressure comes from warm rising air (which forms clouds and rain), while high pressure comes from cool sinking air (which clears the skies).

? Quick Check 4.5

1. What is a front?
2. Which kind of front usually brings sudden, strong storms?
3. What two ingredients does a thunderstorm need?

4.6 Hurricanes

A **hurricane** is the largest and most powerful storm on Earth. It is a giant, spinning system of thunderstorms — and it gets all its energy from one source: **warm ocean water**.

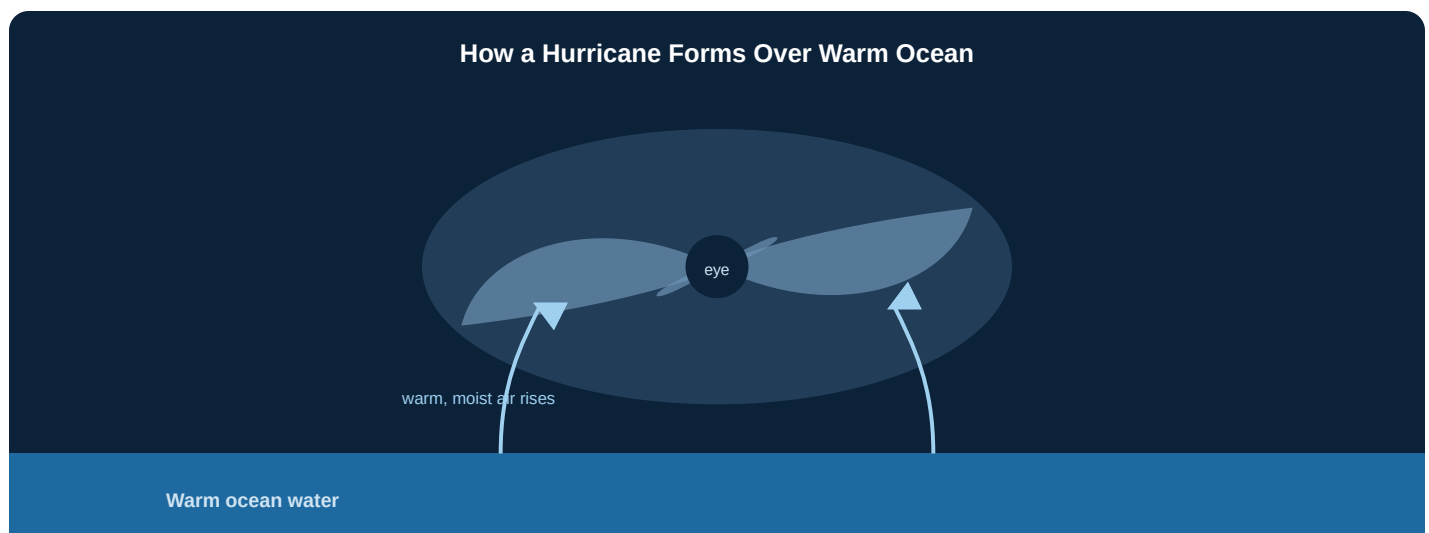


Figure 4.5 — Hurricanes form over warm oceans. Water evaporates, rises, and releases energy that powers the storm. The calm center is called the "eye."

Here's how it works: the Sun heats the ocean, and warm water **evaporates** into the air. As this warm, moist air rises and cools, it releases huge amounts of energy, which feeds the storm and makes it grow. This is why hurricanes form over warm tropical oceans, get stronger over water, and weaken once they move over land (where they lose their warm-water fuel). It also connects back to Chapter 3 — evaporation from the ocean is the fuel.

★ The Ocean Is the Fuel Tank

The warmer the ocean, the more moisture evaporates, and the more energy is available to power the storm. Hurricanes are basically machines that turn warm ocean water into wind and rain.

? Quick Check 4.6

1. Where do hurricanes get their energy?
2. Why do hurricanes weaken after they move over land?
3. What is the calm center of a hurricane called?

Chapter 4 Wrap-Up

★ Key Takeaways

- **Weather** is right now; **climate** is the long-term average.
- The **atmosphere** has layers; we live in the troposphere. The ozone layer blocks UV; greenhouse gases trap heat.
- **Uneven heating** (land vs. water, equator vs. poles) is the starting point for all weather.
- **Wind** forms when warm air rises and cooler air flows in; it moves from high to low pressure.
- **Fronts** form where air masses meet and cause storms; **hurricanes** are powered by warm ocean water.